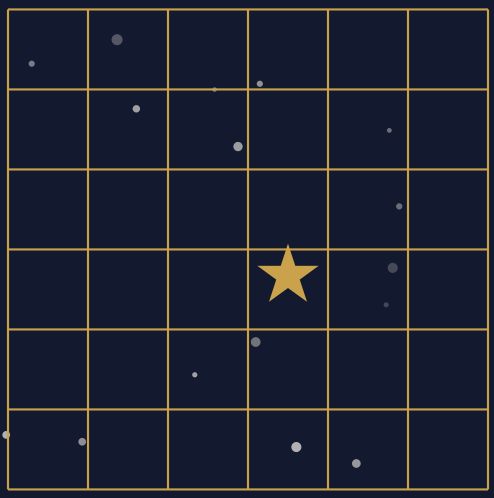
A 24-DAY MURDER MYSTERY ADVENT CALENDAR

THE MIDWINTER THEOREM

36 suspects · 24 logic puzzles · 1 killer



Solve one puzzle a day from December 1 to 24.
Each answer reveals one true clue. Find the killer on Christmas Eve.

MathRelaxing

How to Play

read this first

The case — Professor Ambrose Vale has been murdered at the Midwinter Gala. Thirty-six guests are snowed in at Ashcombe Hall. One of them is the killer.

Each day — Open the page for today's date. Read the journal entry, then solve the logic puzzle. Every puzzle has exactly one solution, so no guessing is needed.

The starred cell — When the puzzle is solved, look at the number in the starred cell. That number tells you which of the day's statements is TRUE. The others are rumours.

Eliminate — Use the true statement to cross suspects off the Suspect Tracker. Statements are always about the culprit, using the facts listed on the Cast pages.

Self-check — If a 'true' statement would eliminate every remaining suspect, your puzzle answer is wrong. Recheck the grid.

Christmas Eve — After Day 24 exactly one suspect remains. Write the name down, then open the sealed Final Reveal. Full puzzle solutions are at the back.

Puzzle types: Sudoku, Futoshiki and Skyscrapers, growing from gentle 4×4 grids to tricky 6×6 and 5×5 grids by the final week. Rules are printed on every page. Play alone, as a couple or as a family. Print double-sided or one day per sheet.

The Cast

suspects 1–18

1 Dr. Beatrix Holm
Field: Statistics
Midnight: Billiard Room
Carrying: Bow Tie
Drinking: Earl Grey
Arrived: 9:30

2 Mr. Oliver Crane
Field: Algebra
Midnight: Ballroom
Carrying: Pocket Watch
Drinking: Earl Grey
Arrived: 8:00

3 Dr. Matilda Fenn
Field: Number Theory
Midnight: Ballroom
Carrying: Scarf
Drinking: Hot Cocoa
Arrived: 7:00

4 Mrs. Constance Reed
Field: Number Theory
Midnight: Ballroom
Carrying: Spectacles
Drinking: Hot Cocoa
Arrived: 8:30

5 Mr. Percival Oates
Field: Analysis
Midnight: Observatory
Carrying: Spectacles
Drinking: Black Coffee
Arrived: 9:30

6 Dr. Nadia Okafor
Field: Statistics
Midnight: Library
Carrying: Spectacles
Drinking: Mulled Wine
Arrived: 7:30

7 Miss Philippa Shaw
Field: Topology
Midnight: Conservatory
Carrying: Pocket Watch
Drinking: Mulled Wine
Arrived: 7:30

8 Mr. Jasper Thorne
Field: Topology
Midnight: Chapel
Carrying: Walking Cane
Drinking: Hot Cocoa
Arrived: 7:00

9 Dr. Agnes Pemberton
Field: Statistics
Midnight: Billiard Room
Carrying: Pocket Watch
Drinking: Black Coffee
Arrived: 8:00

10 Mrs. Lydia Fairfax
Field: Topology
Midnight: Library
Carrying: Bow Tie
Drinking: Mulled Wine
Arrived: 9:30

11 Mrs. Ottilie Grey
Field: Geometry
Midnight: Conservatory
Carrying: Bow Tie
Drinking: Hot Cocoa
Arrived: 8:00

12 Dr. Helena Moreau
Field: Statistics
Midnight: Ballroom
Carrying: Scarf
Drinking: Mulled Wine
Arrived: 7:00

13 Prof. Ignatius Lowe
Field: Number Theory
Midnight: Ballroom
Carrying: Pocket Watch
Drinking: Black Coffee
Arrived: 7:00

14 Prof. Rupert Ashby
Field: Algebra
Midnight: Conservatory
Carrying: Pocket Watch
Drinking: Black Coffee
Arrived: 8:00

15 Prof. Hugo Lindqvist
Field: Algebra
Midnight: Billiard Room
Carrying: Fountain Pen
Drinking: Hot Cocoa
Arrived: 8:30

16 Mr. Lucian Hargrove
Field: Number Theory
Midnight: Chapel
Carrying: Spectacles
Drinking: Hot Cocoa
Arrived: 9:30

17 Prof. Leopold Brandt
Field: Number Theory
Midnight: Observatory
Carrying: Walking Cane
Drinking: Hot Cocoa
Arrived: 9:30

18 Dr. Margot Ellery
Field: Analysis
Midnight: Billiard Room
Carrying: Spectacles
Drinking: Mulled Wine
Arrived: 9:00

The Cast

suspects 19–36

19 Dr. Estelle Park
Field: Number Theory
Midnight: Billiard Room
Carrying: Fountain Pen
Drinking: Hot Cocoa
Arrived: 9:00

20 Miss Josephine Lark
Field: Number Theory
Midnight: Chapel
Carrying: Bow Tie
Drinking: Earl Grey
Arrived: 7:00

21 Miss Rosalind Kerr
Field: Topology
Midnight: Observatory
Carrying: Scarf
Drinking: Earl Grey
Arrived: 8:00

22 Prof. Cornelius Bell
Field: Analysis
Midnight: Ballroom
Carrying: Spectacles
Drinking: Hot Cocoa
Arrived: 7:30

23 Miss Harriet Vance
Field: Number Theory
Midnight: Billiard Room
Carrying: Walking Cane
Drinking: Hot Cocoa
Arrived: 7:00

24 Prof. Ansel Marchetti
Field: Geometry
Midnight: Billiard Room
Carrying: Walking Cane
Drinking: Earl Grey
Arrived: 7:00

25 Mrs. Eloise Tran
Field: Number Theory
Midnight: Ballroom
Carrying: Bow Tie
Drinking: Earl Grey
Arrived: 9:00

26 Mr. Barnaby Fitch
Field: Statistics
Midnight: Chapel
Carrying: Bow Tie
Drinking: Black Coffee
Arrived: 9:00

27 Mr. Felix Dunmore
Field: Geometry
Midnight: Billiard Room
Carrying: Scarf
Drinking: Hot Cocoa
Arrived: 8:30

28 Prof. Silas Wren
Field: Geometry
Midnight: Observatory
Carrying: Pocket Watch
Drinking: Hot Cocoa
Arrived: 8:00

29 Miss Clara Winslow
Field: Topology
Midnight: Library
Carrying: Spectacles
Drinking: Hot Cocoa
Arrived: 9:30

30 Prof. Victor Abernathy
Field: Algebra
Midnight: Chapel
Carrying: Bow Tie
Drinking: Black Coffee
Arrived: 7:30

31 Mr. Gideon Pratt
Field: Number Theory
Midnight: Library
Carrying: Fountain Pen
Drinking: Hot Cocoa
Arrived: 7:00

32 Mr. Theodore Quill
Field: Number Theory
Midnight: Billiard Room
Carrying: Bow Tie
Drinking: Mulled Wine
Arrived: 7:30

33 Dr. Iris Calloway
Field: Geometry
Midnight: Observatory
Carrying: Bow Tie
Drinking: Mulled Wine
Arrived: 8:00

34 Mr. Ambrose Kettle
Field: Geometry
Midnight: Ballroom
Carrying: Bow Tie
Drinking: Black Coffee
Arrived: 9:00

35 Prof. Edmund Hale
Field: Algebra
Midnight: Billiard Room
Carrying: Bow Tie
Drinking: Mulled Wine
Arrived: 9:00

36 Dr. Vivian Sato
Field: Geometry
Midnight: Library
Carrying: Fountain Pen
Drinking: Earl Grey
Arrived: 7:00

Suspect Tracker

cross off each suspect the day they are eliminated

- ☐ 1. Dr. Beatrix Holm — day ___
- ☐ 2. Mr. Oliver Crane — day ___
- ☐ 3. Dr. Matilda Fenn — day ___
- ☐ 4. Mrs. Constance Reed — day ___
- ☐ 5. Mr. Percival Oates — day ___
- ☐ 6. Dr. Nadia Okafor — day ___
- ☐ 7. Miss Philippa Shaw — day ___
- ☐ 8. Mr. Jasper Thorne — day ___
- ☐ 9. Dr. Agnes Pemberton — day ___
- ☐ 10. Mrs. Lydia Fairfax — day ___
- ☐ 11. Mrs. Ottilie Grey — day ___
- ☐ 12. Dr. Helena Moreau — day ___
- ☐ 13. Prof. Ignatius Lowe — day ___
- ☐ 14. Prof. Rupert Ashby — day ___
- ☐ 15. Prof. Hugo Lindqvist — day ___
- ☐ 16. Mr. Lucian Hargrove — day ___
- ☐ 17. Prof. Leopold Brandt — day ___
- ☐ 18. Dr. Margot Ellery — day ___
- ☐ 19. Dr. Estelle Park — day ___
- ☐ 20. Miss Josephine Lark — day ___
- ☐ 21. Miss Rosalind Kerr — day ___
- ☐ 22. Prof. Cornelius Bell — day ___
- ☐ 23. Miss Harriet Vance — day ___
- ☐ 24. Prof. Ansel Marchetti — day ___
- ☐ 25. Mrs. Eloise Tran — day ___
- ☐ 26. Mr. Barnaby Fitch — day ___
- ☐ 27. Mr. Felix Dunmore — day ___
- ☐ 28. Prof. Silas Wren — day ___
- ☐ 29. Miss Clara Winslow — day ___
- ☐ 30. Prof. Victor Abernathy — day ___
- ☐ 31. Mr. Gideon Pratt — day ___
- ☐ 32. Mr. Theodore Quill — day ___
- ☐ 33. Dr. Iris Calloway — day ___
- ☐ 34. Mr. Ambrose Kettle — day ___
- ☐ 35. Prof. Edmund Hale — day ___
- ☐ 36. Dr. Vivian Sato — day ___

December 1

day 1 of 24

Snow sealed the roads to Ashcombe Hall on the night of the Midwinter Gala. At midnight the Society's president, Professor Ambrose Vale, was found dead beside his notebook, the final page torn out. He had promised to reveal a proof of the Midwinter Conjecture at dawn.

Today's puzzle: Sudoku 4×4

Fill the grid so every row, column and outlined box contains each number exactly once.

	4	2	★
2		4	1
	2	1	
	1		

The starred number tells you which statement is true:

1. The culprit works in either Algebra or Analysis.
2. The culprit works in either Number Theory or Statistics.
3. The culprit was not in the Library at midnight.
4. The culprit arrived at 9:30 or later.

NOTES · suspects eliminated today:

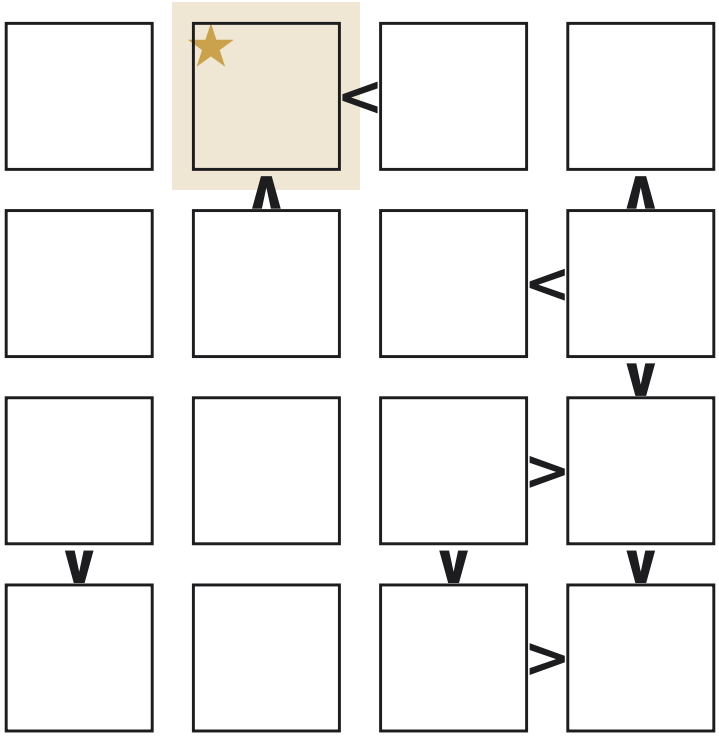
December 2

day 2 of 24

The butler confirms the doors were bolted from eleven o'clock. Whoever silenced Vale never left the Hall. Thirty-six guests remain snowed in, and every one of them had heard of the proof.

Today's puzzle: Futoshiki 4×4

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. The culprit's first name has more than five letters.
2. Nobody drinking Hot Cocoa in the Billiard Room is guilty.
3. The culprit was not drinking Hot Cocoa.
4. If the culprit drank Hot Cocoa, they arrived before 7:30.

NOTES · suspects eliminated today:

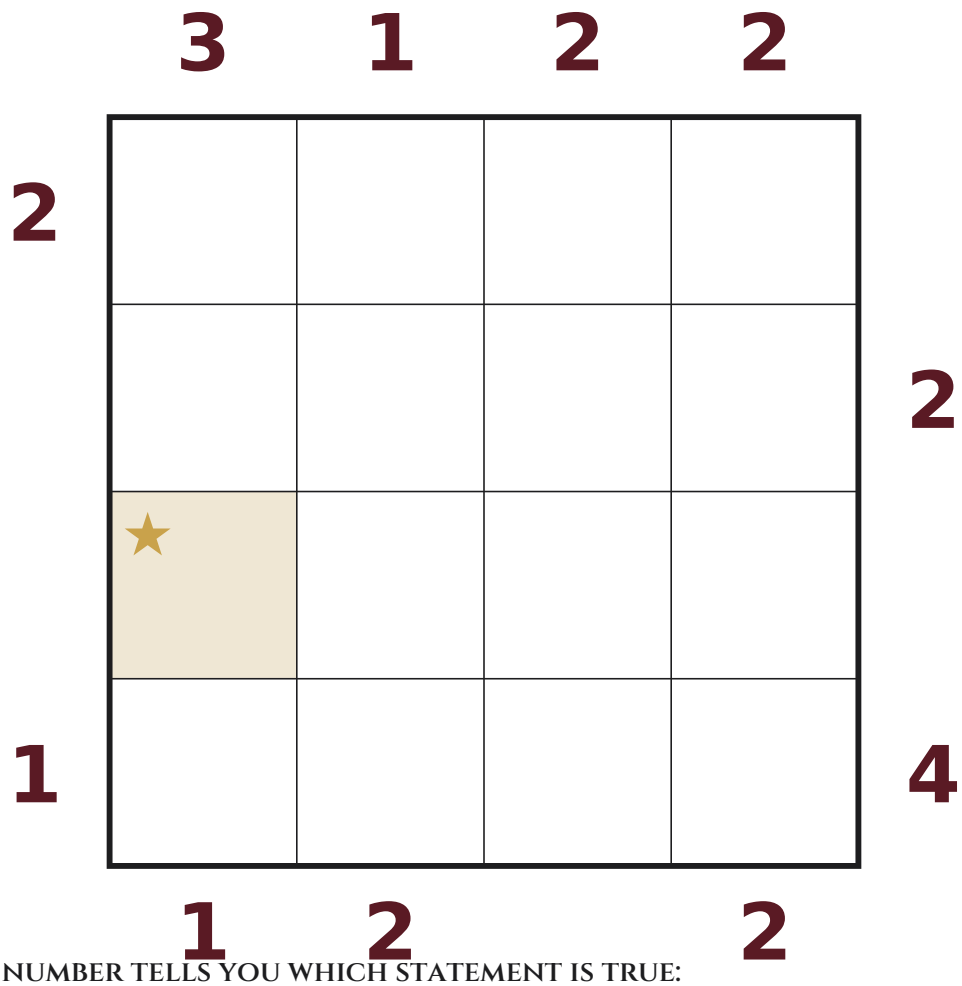
December 3

day 3 of 24

Vale's teacup was still warm when he was found. The kitchen kept a careful record of what each guest drank that evening, and the housekeeper is willing to talk.

Today's puzzle: Skyscrapers 4×4

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. The culprit does not work in Statistics.
2. At midnight the culprit was in either the Library or the Ballroom.
3. The culprit works in either Algebra or Statistics.
4. At midnight the culprit was in either the Ballroom or the Conservatory.

NOTES · suspects eliminated today:

December 4

day 4 of 24

A trail of wet footprints crosses the marble floor and fades near the grand staircase. Someone came in from the cold not long before midnight.

Today's puzzle: Sudoku 4×4

Fill the grid so every row, column and outlined box contains each number exactly once.

4		★	
2			1
	2		4
1	4		2

The starred number tells you which statement is true:

1. At midnight the culprit was in either the Ballroom or the Chapel.
2. If the culprit drank Black Coffee, they arrived before 8:30.
3. The culprit was not in the Observatory at midnight.
4. At midnight the culprit was in either the Library or the Conservatory.

NOTES · suspects eliminated today:

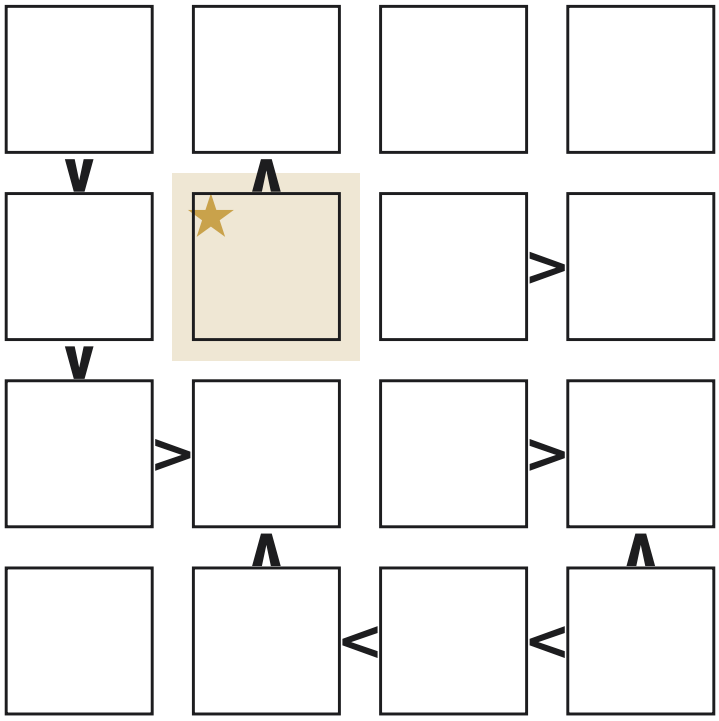
December 5

day 5 of 24

In the margins of Vale's notebook: a column of crossed-out names and a single underlined word, 'betrayal'. He suspected someone close to him.

Today's puzzle: Futoshiki 4×4

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. The culprit is not a professor.
2. The culprit arrived before 8:00.
3. At midnight the culprit was in either the Billiard Room or the Chapel.
4. The culprit is not a Topology scholar carrying a pocket watch.

NOTES · suspects eliminated today:

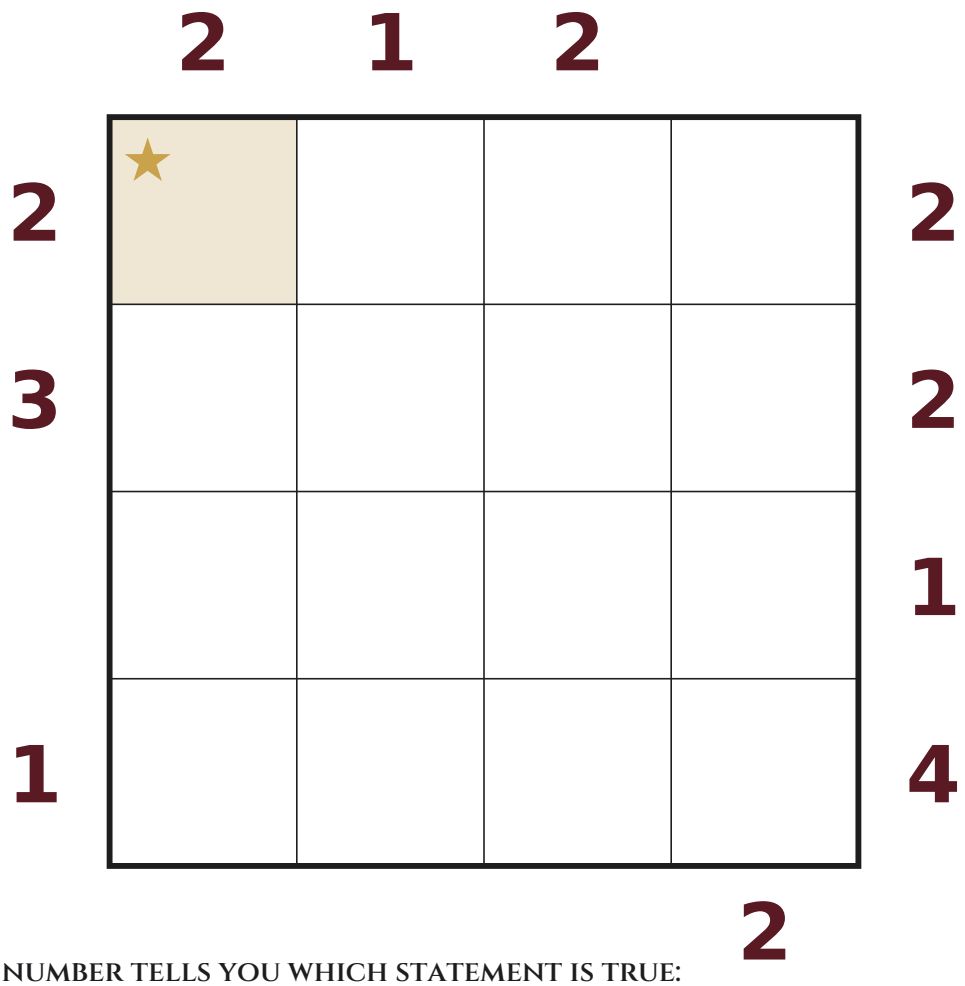
December 6

day 6 of 24

The guest book is water-stained but legible. Arrival times can be trusted, the porter insists; he wrote each one down himself.

Today's puzzle: Skyscrapers 4×4

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. If the culprit drank Hot Cocoa, they arrived before 8:00.
2. The culprit works in either Analysis or Topology.
3. Nobody drinking Earl Grey in the Observatory is guilty.
4. At midnight the culprit was in either the Conservatory or the Billiard Room.

NOTES · suspects eliminated today:

December 7

day 7 of 24

A cracked pocket mirror lies under the piano. The pianist swears it was not there during the waltz.

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

	6	4			3
	2	★		5	
			4		
2	4			3	
4	5	1	3	2	
	3	2	5	4	

The starred number tells you which statement is true:

1. The culprit works in either Algebra or Topology.
2. At midnight the culprit was in either the Library or the Billiard Room.
3. The culprit is not a Topology scholar carrying a walking cane.
4. The culprit arrived before 7:30.
5. The culprit is not a Geometry scholar carrying a pocket watch.
6. The culprit arrived at 8:30 or later.

NOTES · suspects eliminated today:

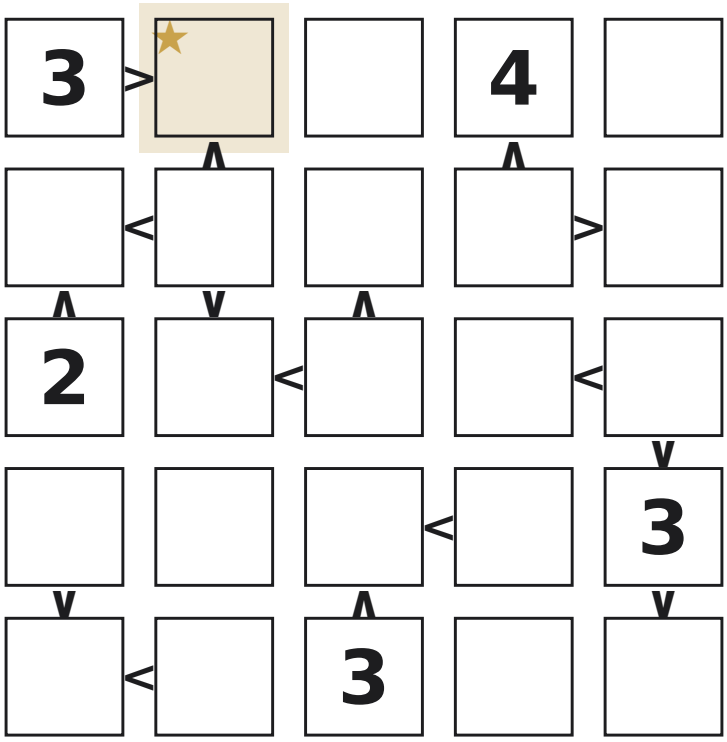
December 8

day 8 of 24

The torn page left an impression on the sheet below. Rubbed with pencil, it shows a lattice of numbers, the start of the proof.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. The culprit works in either Analysis or Statistics.
2. If the culprit drank Black Coffee, they arrived before 8:00.
3. Nobody drinking Hot Cocoa in the Observatory is guilty.
4. The culprit works in either Topology or Number Theory.
5. At midnight the culprit was in either the Ballroom or the Billiard Room.

NOTES · suspects eliminated today:

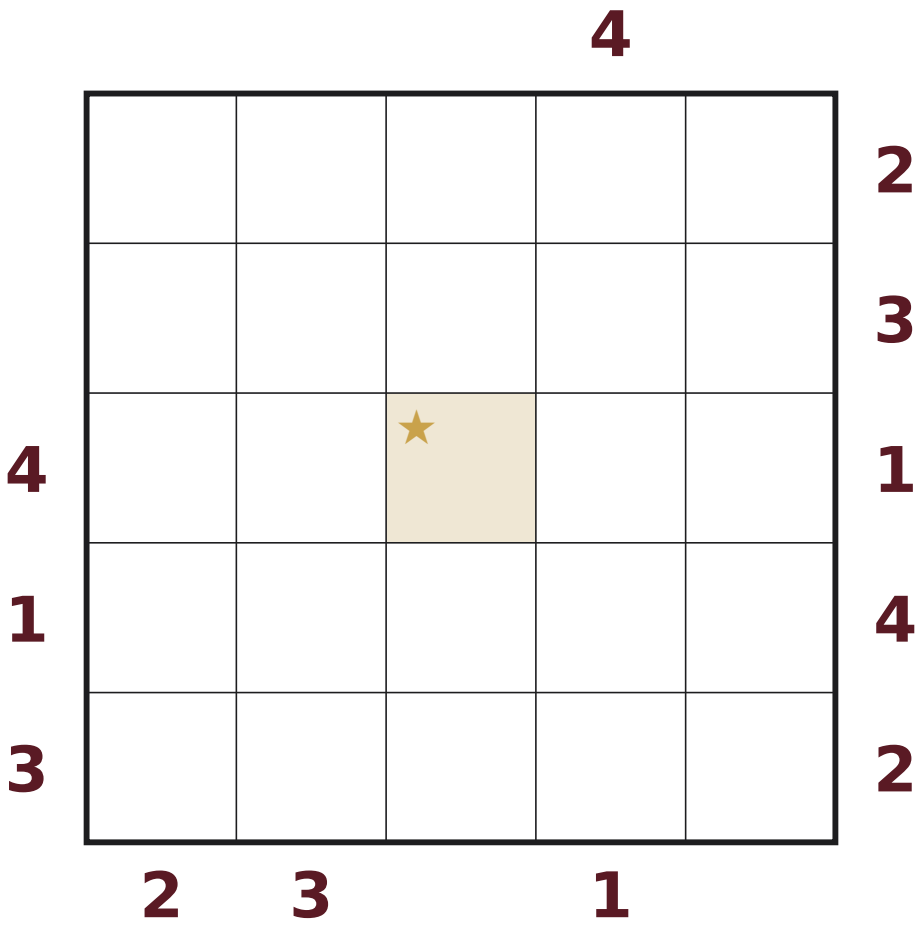
December 9

day 9 of 24

Two guests quarrelled loudly in the corridor at half past ten. Neither will say what about.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. At midnight the culprit was in either the Library or the Chapel.
2. Nobody drinking Earl Grey in the Billiard Room is guilty.
3. At midnight the culprit was in either the Conservatory or the Chapel.
4. The culprit works in either Algebra or Number Theory.
5. The culprit works in either Topology or Statistics.

NOTES · suspects eliminated today:

December 10

day 10 of 24

The observatory telescope was turned away from the stars and toward the east wing windows. Someone was watching the Hall rather than the sky.

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

5		4		6	
6	1				5
	6		5		
4				2	6
3		5	★		4
1	4			5	2

The starred number tells you which statement is true:

1. The culprit was not carrying a pocket watch.
2. The culprit works in either Analysis or Number Theory.
3. The culprit does not work in Geometry.
4. The culprit arrived at 9:00 or later.
5. At midnight the culprit was in either the Library or the Chapel.
6. The culprit is not a Algebra scholar carrying a pocket watch.

NOTES · suspects eliminated today:

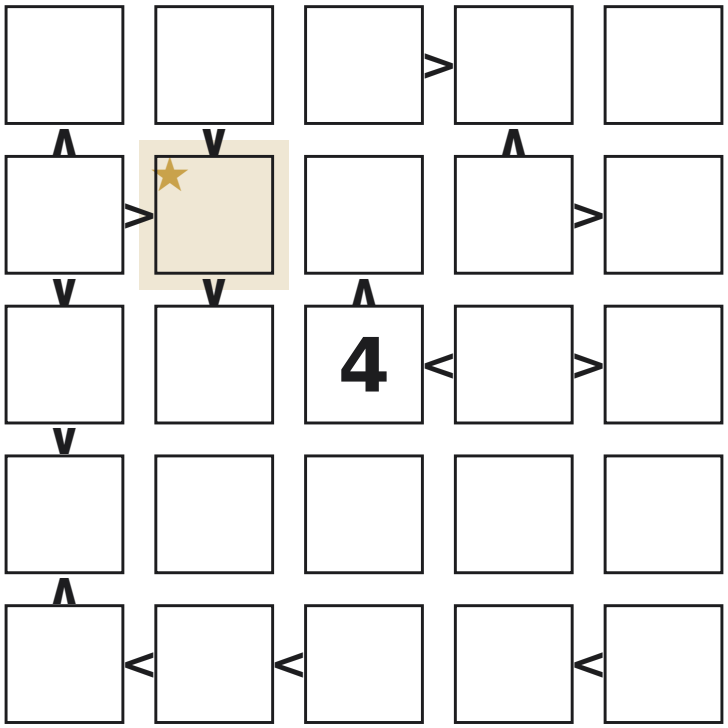
December 11

day 11 of 24

Vale's will was amended last week. The solicitor, stranded with everyone else, is nervous.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. The culprit arrived at 9:00 or later.
2. If the culprit drank Black Coffee, they arrived before 7:30.
3. At midnight the culprit was in either the Ballroom or the Conservatory.
4. At midnight the culprit was in either the Conservatory or the Billiard Room.
5. At midnight the culprit was in either the Library or the Ballroom.

NOTES · suspects eliminated today:

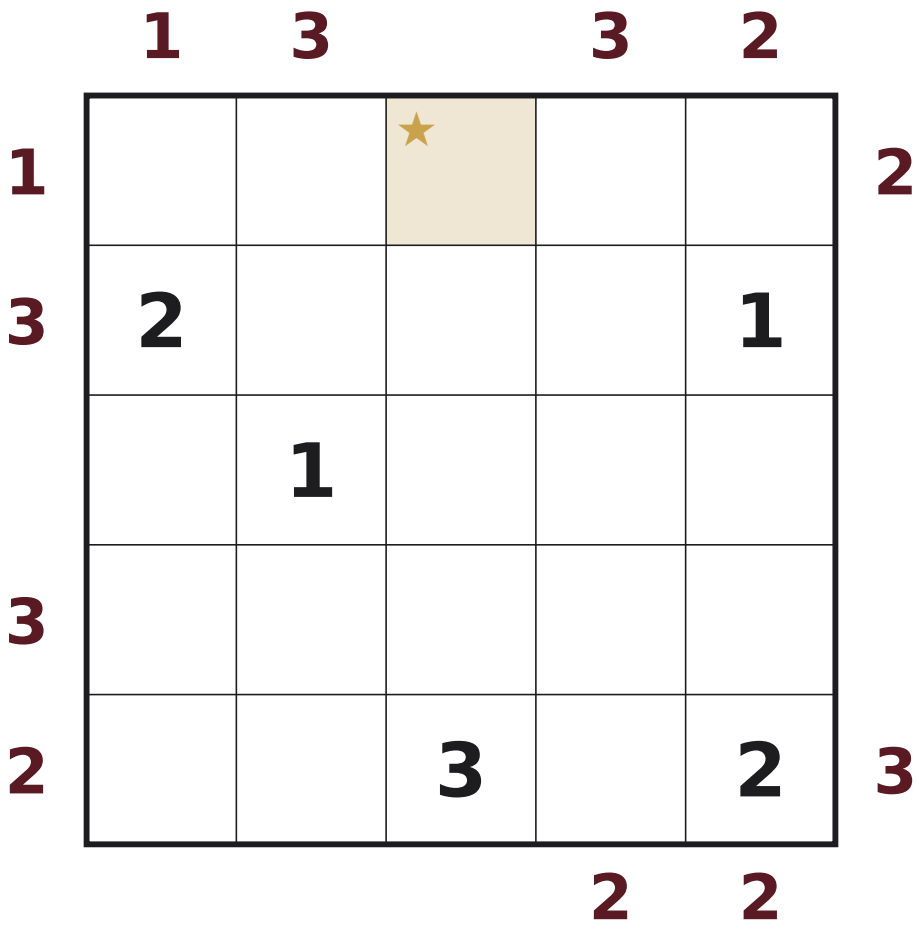
December 12

day 12 of 24

The chapel candles were relit after the evening service. Wax drips on the floor are still soft.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. The culprit is not a Number Theory scholar carrying a walking cane.
2. The culprit does not work in Geometry.
3. The culprit arrived at 9:30 or later.
4. The culprit is not a professor.
5. At midnight the culprit was in either the Billiard Room or the Chapel.

NOTES · suspects eliminated today:

December 13

day 13 of 24

A frost-rimed window in the conservatory has been forced from the inside, but the snow beneath it is untouched. A false trail.

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

2			4	5	6
6	5	4	2		
5					3
1	6	3	5		
	1			6	
	2	6			★

The starred number tells you which statement is true:

1. The culprit works in either Algebra or Number Theory.
2. The culprit works in either Analysis or Number Theory.
3. The culprit is not a Geometry scholar carrying a pocket watch.
4. The culprit works in either Algebra or Statistics.
5. Nobody drinking Hot Cocoa in the Conservatory is guilty.
6. At midnight the culprit was in either the Conservatory or the Chapel.

NOTES · suspects eliminated today:

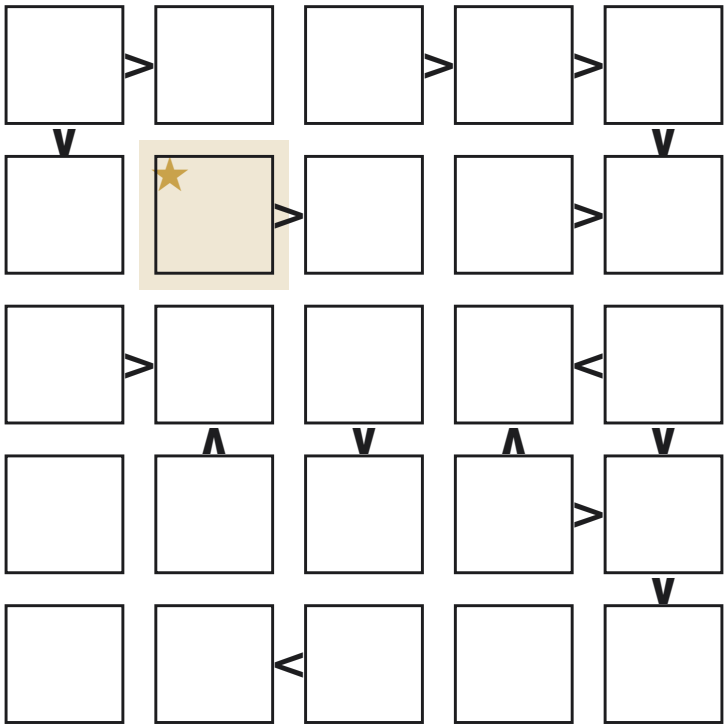
December 14

day 14 of 24

The housekeeper recalls a guest asking, twice, where the president would be sitting at midnight.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. The culprit works in either Topology or Number Theory.
2. The culprit was not drinking Hot Cocoa.
3. Nobody drinking Hot Cocoa in the Observatory is guilty.
4. The culprit arrived at 8:30 or later.
5. The culprit was not drinking Black Coffee.

NOTES · suspects eliminated today:

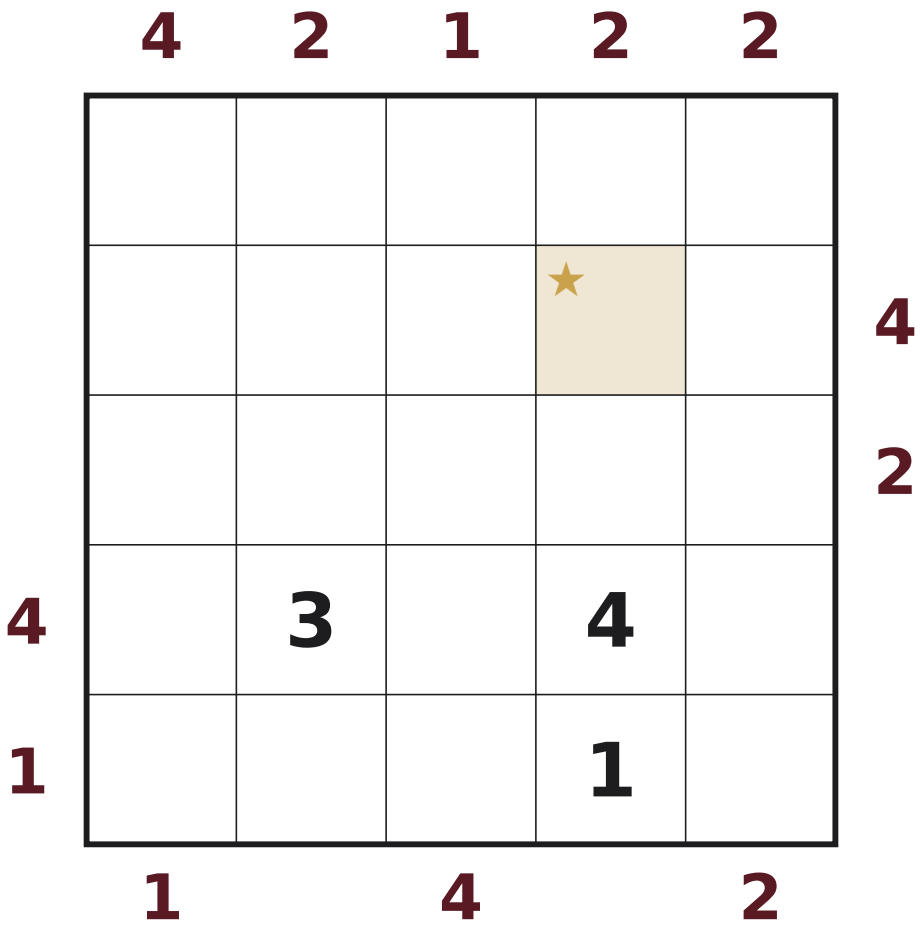
December 15

day 15 of 24

Among the billiard balls one is missing. It turns up later, wrapped in a handkerchief, in a coat pocket.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. At midnight the culprit was in either the Library or the Conservatory.
2. Nobody drinking Hot Cocoa in the Chapel is guilty.
3. If the culprit drank Hot Cocoa, they arrived before 8:00.
4. The culprit was not carrying a pocket watch.
5. The culprit's first name has more than five letters.

NOTES · suspects eliminated today:

December 16

day 16 of 24

A letter in Vale's desk, unsent: 'If anything should happen to me, look to the one who counts the hours.'

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

	2			1	3
		3		6	5
			3		
	1			5	
5		★	1		4
	3				6

The starred number tells you which statement is true:

1. The culprit was not in the Observatory at midnight.
2. The culprit was not in the Chapel at midnight.
3. The culprit works in either Number Theory or Statistics.
4. The culprit arrived before 8:00.
5. The culprit works in either Analysis or Topology.
6. The culprit arrived before 7:30.

NOTES · suspects eliminated today:

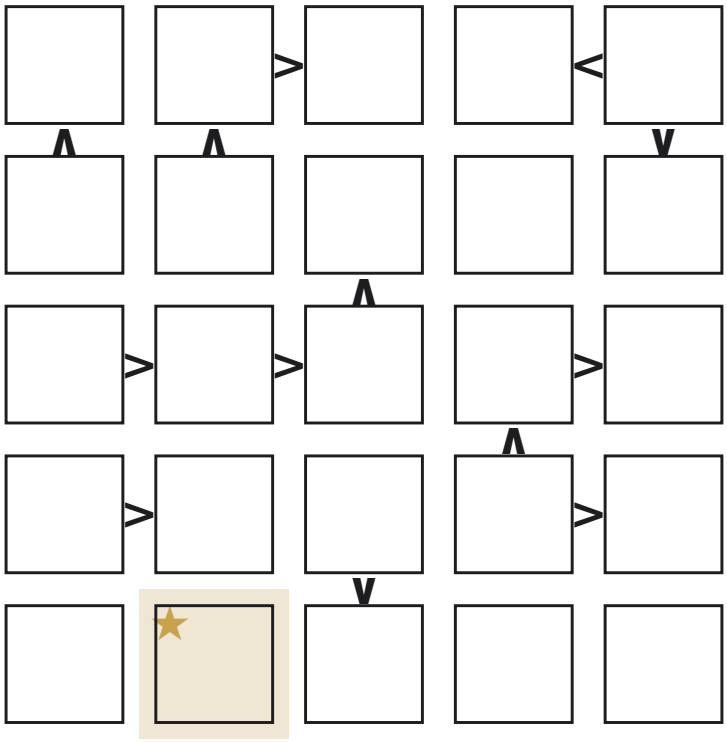
December 17

day 17 of 24

The library ledger shows a rare volume on lattice theory borrowed at nine and never returned.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. At midnight the culprit was in either the Ballroom or the Chapel.
2. Nobody drinking Earl Grey in the Ballroom is guilty.
3. The culprit works in either Algebra or Topology.
4. If the culprit drank Hot Cocoa, they arrived before 7:30.
5. The culprit works in either Analysis or Statistics.

NOTES · suspects eliminated today:

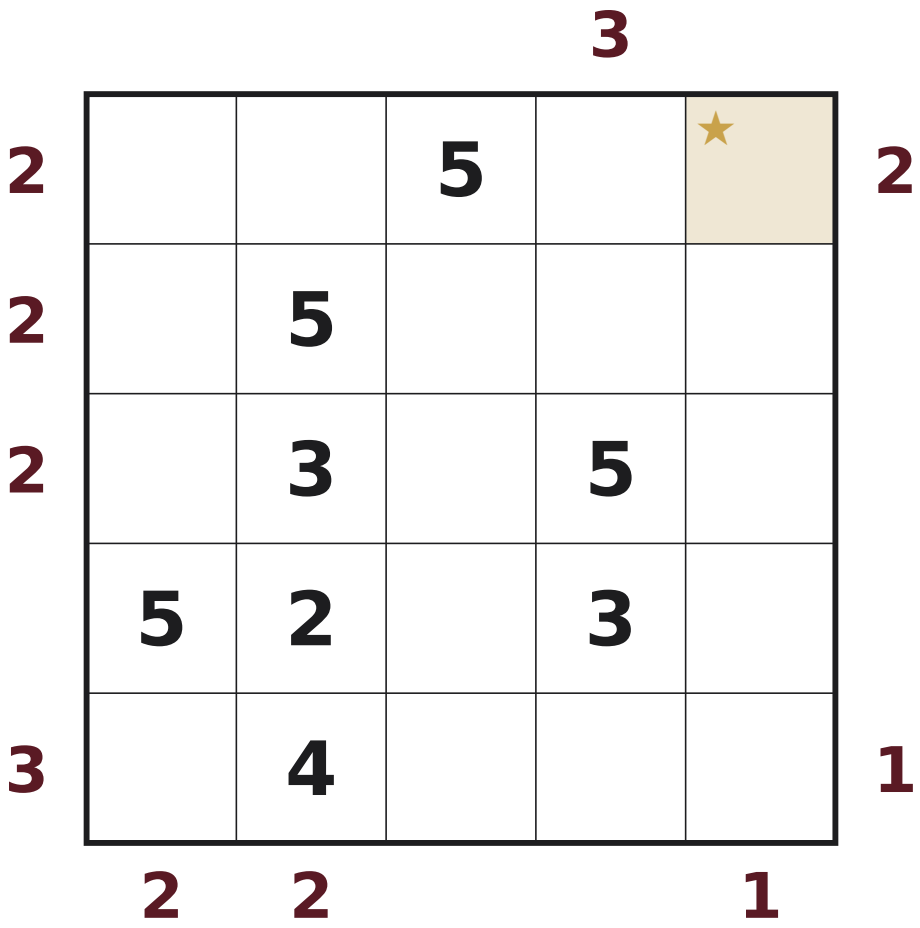
December 18

day 18 of 24

Half the guests have gone to bed. The other half are watching each other over cold coffee. The suspect list is shrinking.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. The culprit works in either Algebra or Analysis.
2. At midnight the culprit was in either the Ballroom or the Billiard Room.
3. At midnight the culprit was in either the Library or the Billiard Room.
4. The culprit is not a Number Theory scholar carrying a spectacles.
5. The culprit works in either Topology or Statistics.

NOTES · suspects eliminated today:

December 19

day 19 of 24

The torn page is found at last, burned to a curl in the ballroom grate. Only a few symbols survive.

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

	4		6		
	3	2		1	
		4	2	3	
3			5		★
	1		3	5	
					2

The starred number tells you which statement is true:

1. The culprit arrived before 7:30.
2. The culprit's first name has more than five letters.
3. At midnight the culprit was in either the Library or the Conservatory.
4. Nobody drinking Hot Cocoa in the Observatory is guilty.
5. At midnight the culprit was in either the Billiard Room or the Chapel.
6. The culprit is not a Number Theory scholar carrying a scarf.

NOTES · suspects eliminated today:

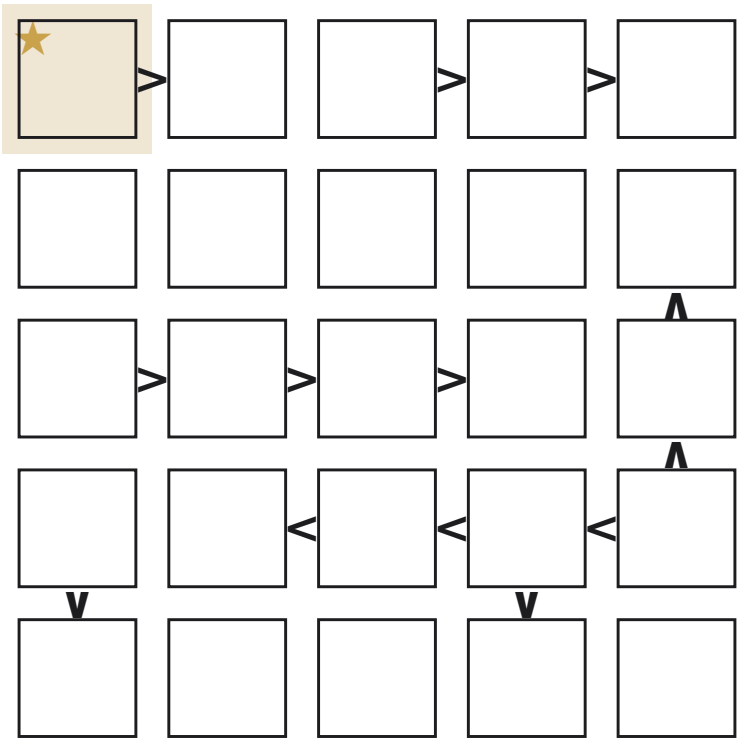
December 20

day 20 of 24

A guest tries to leave on foot through the snow and is brought back, shivering, by the gardener.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. If the culprit drank Hot Cocoa, they arrived before 7:30.
2. The culprit arrived at 8:30 or later.
3. The culprit does not work in Geometry.
4. Nobody drinking Mulled Wine in the Observatory is guilty.
5. At midnight the culprit was in either the Library or the Ballroom.

NOTES · suspects eliminated today:

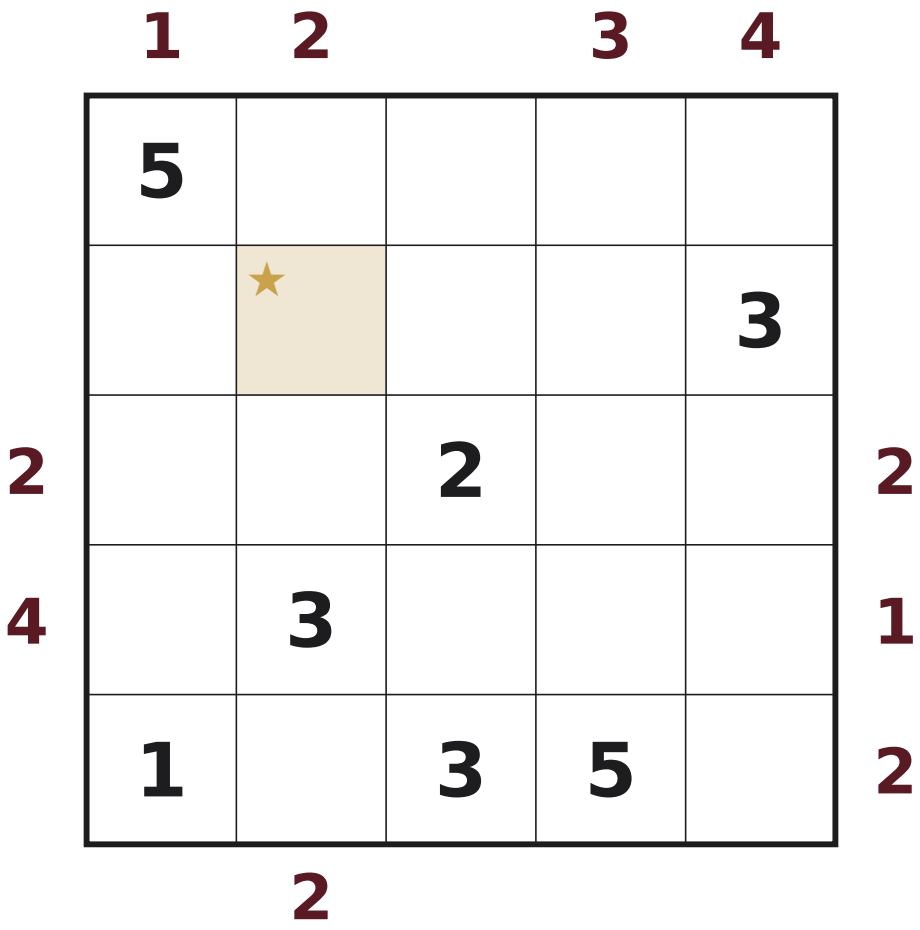
December 21

day 21 of 24

The pattern is clear now. The killer knew the proof was real and wanted the credit.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. The culprit is not a doctor.
2. The culprit works in either Algebra or Topology.
3. The culprit works in either Analysis or Statistics.
4. The culprit works in either Number Theory or Statistics.
5. At midnight the culprit was in either the Library or the Billiard Room.

NOTES · suspects eliminated today:

December 22

day 22 of 24

Only a handful of names remain on your list. Each of them is watching the door.

Today's puzzle: Sudoku 6×6

Fill the grid so every row, column and outlined box contains each number exactly once.

		4			
	1	5	2		
	★			5	
			3		
					2
2		1	4		3

The starred number tells you which statement is true:

1. The culprit was not in the Observatory at midnight.
2. At midnight the culprit was in either the Library or the Chapel.
3. At midnight the culprit was in either the Observatory or the Billiard Room.
4. The culprit works in either Analysis or Number Theory.
5. At midnight the culprit was in either the Ballroom or the Chapel.
6. The culprit works in either Topology or Number Theory.

NOTES · suspects eliminated today:

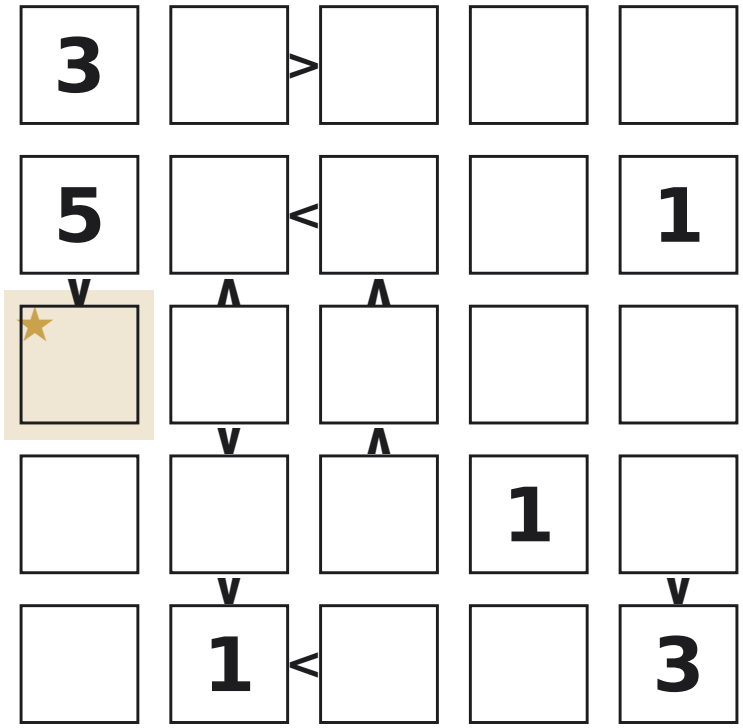
December 23

day 23 of 24

On the eve of Christmas Eve, the last few alibis are about to crumble. Tomorrow you will know.

Today's puzzle: Futoshiki 5×5

Fill the grid so every row and column contains each number once. Every inequality sign between two cells, across or down, must be obeyed: it always points at the smaller number.



The starred number tells you which statement is true:

1. If the culprit drank Mulled Wine, they arrived before 8:30.
2. The culprit works in either Algebra or Statistics.
3. The culprit works in either Analysis or Topology.
4. The culprit works in either Topology or Statistics.
5. At midnight the culprit was in either the Ballroom or the Conservatory.

NOTES · suspects eliminated today:

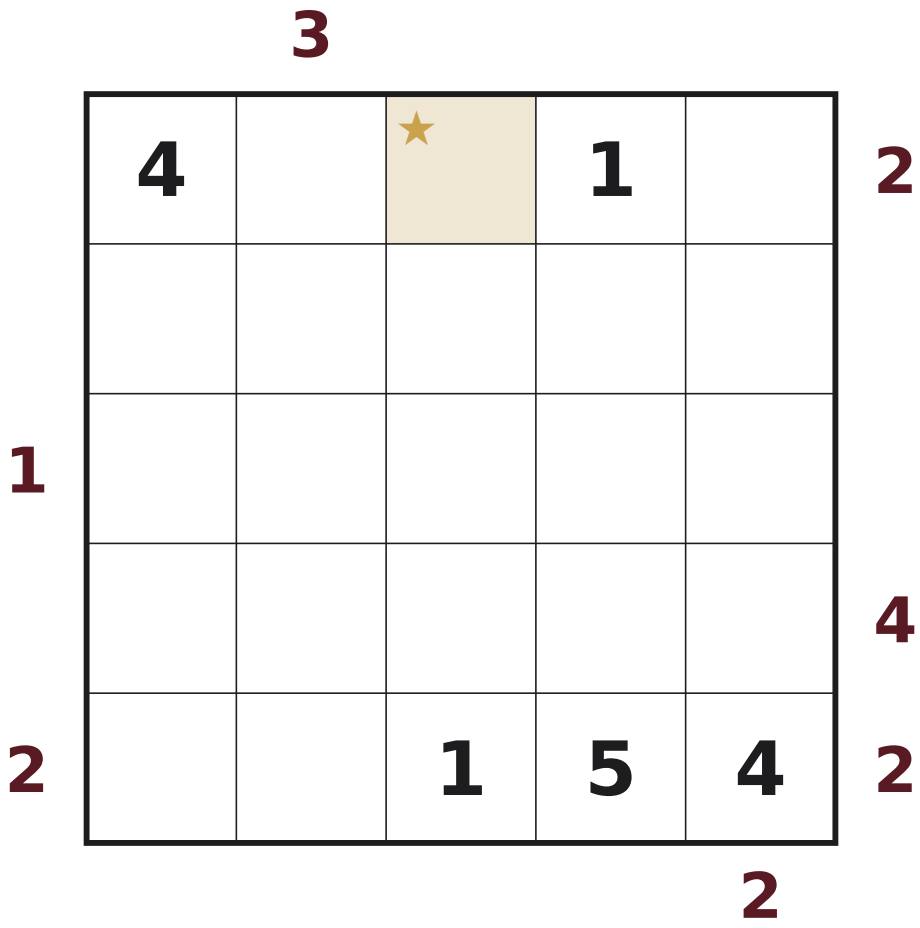
December 24

day 24 of 24

One final clue. When this puzzle is solved, exactly one name will remain on your list. Write it down, then turn to the sealed Final Reveal.

Today's puzzle: Skyscrapers 5×5

Each number is a building of that height; every row and column holds each height once. A number outside the grid says how many buildings you can see from that side (taller ones hide shorter ones behind them).



The starred number tells you which statement is true:

1. The culprit is not a professor.
2. The culprit arrived at 9:00 or later.
3. The culprit arrived at 9:30 or later.
4. The culprit works in either Algebra or Number Theory.
5. If the culprit drank Mulled Wine, they arrived before 7:30.

NOTES · suspects eliminated today:

Final Reveal

do not read before solving Day 24

Fold this page in half along the dotted line and seal it until Christmas Eve.

Case closed.

The killer is suspect 28, Prof. Silas Wren, a specialist in Geometry. Arriving at 8:00 with a pocket watch and a glass of Hot Cocoa, Wren slipped away to the Observatory just before midnight. Vale had shown them the proof in confidence, and they could not bear to see him take the credit. The torn page was meant to be published under their own name. They had not reckoned on a detective who could count.

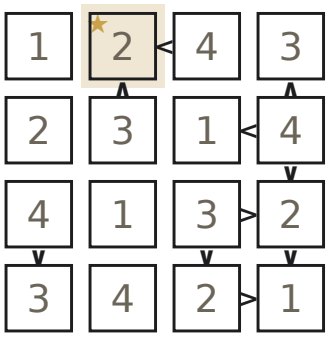
Solutions

days 1–6

Day 1 · star = 3

1	**4**	**2**	★ 3
2	3	**4**	**1**
3	**2**	**1**	4
4	**1**	3	2

Day 2 · star = 2



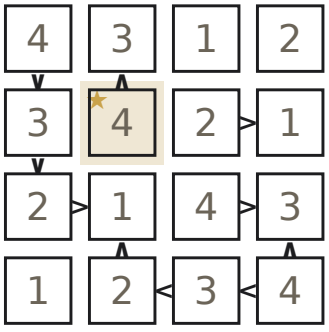
Day 3 · star = 1

	3	**1**	**2**	**2**	
2	2	4	1	3	
	3	1	4	2	**2**
	★ 1	2	3	4	
1	4	3	2	1	**4**
	1	**2**		**2**	

Day 4 · star = 2

4	1	★ 2	3
2	3	4	**1**
3	**2**	1	**4**
1	**4**	3	**2**

Day 5 · star = 4



Day 6 · star = 3

	2	**1**	**2**	
2	★ 3	4	1	2	**2**
3	1	2	4	3	**2**
	2	1	3	4	**1**
1	4	3	2	1	**4**
				2	

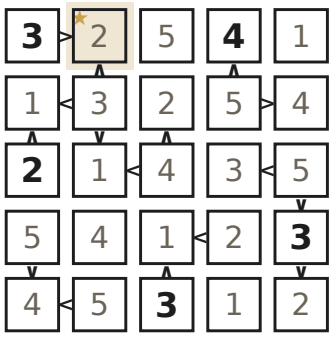
Solutions

days 7–12

Day 7 · star = 3

5	**6**	**4**	2	1	**3**
1	**2**	★3	6	**5**	4
3	1	5	**4**	6	2
2	**4**	6	1	**3**	5
4	**5**	**1**	**3**	**2**	6
6	**3**	**2**	**5**	**4**	1

Day 8 · star = 2



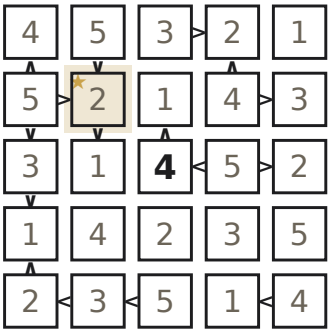
Day 9 · star = 2

			4		
	3	5	1	2	4	2
	4	2	5	3	1	3
4	1	3	★2	4	5	1
1	5	4	3	1	2	4
3	2	1	4	5	3	2
	2	3		1		

Day 10 · star = 6

5	3	**4**	2	**6**	1
6	**1**	2	4	3	**5**
2	**6**	1	**5**	4	3
4	5	3	1	**2**	**6**
3	2	**5**	★6	1	**4**
1	**4**	6	3	**5**	**2**

Day 11 · star = 2



Day 12 · star = 1

	1	3		3	2	
1	5	3	★1	2	4	2
3	**2**	4	5	3	**1**	
	4	**1**	2	5	3	
3	3	2	4	1	5	
2	1	5	**3**	4	**2**	3
				2	**2**	

Solutions

days 13–18

Day 13 · star = 5

2	3	1	**4**	**5**	**6**
6	**5**	**4**	**2**	3	1
5	4	2	6	1	**3**
1	**6**	**3**	**5**	2	4
4	**1**	5	3	**6**	2
3	**2**	**6**	1	4	★5

Day 14 · star = 5

5	1	4	3	2
3	★5	2	4	1
4	2	3	1	5
2	3	1	5	4
1	4	5	2	3

Day 15 · star = 2

	4	**2**	**1**	**2**	**2**
	1	2	5	3	4
	3	5	4	★2	1	**4**
	4	1	3	5	2	**2**
4	2	**3**	1	**4**	5	
1	5	4	2	**1**	3	
	1		**4**		**2**	

Day 16 · star = 2

6	**2**	5	4	**1**	**3**
1	4	**3**	2	**6**	**5**
2	5	6	**3**	4	1
3	**1**	4	6	**5**	2
5	6	★2	**1**	3	**4**
4	**3**	1	5	2	**6**

Day 17 · star = 2

2	3	1	4	5
3	5	2	1	4
5	4	3	2	1
4	1	5	3	2
1	★2	4	5	3

Day 18 · star = 4

			3			
2	3	1	**5**	2	★4	**2**
2	1	**5**	2	4	3	
2	4	**3**	1	**5**	2	
	5	**2**	4	**3**	1	
3	2	**4**	3	1	5	**1**
	2	**2**			**1**	

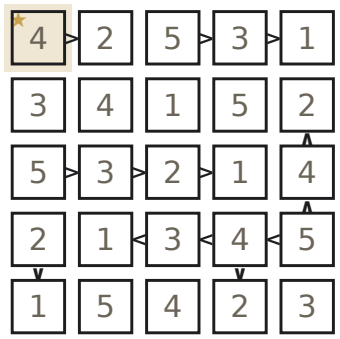
Solutions

days 19–24

Day 19 · star = 6

1	**4**	5	**6**	2	3
6	**3**	**2**	4	**1**	5
5	6	**4**	**2**	**3**	1
3	2	1	**5**	4	★6
2	**1**	6	**3**	**5**	4
4	5	3	1	6	**2**

Day 20 · star = 4



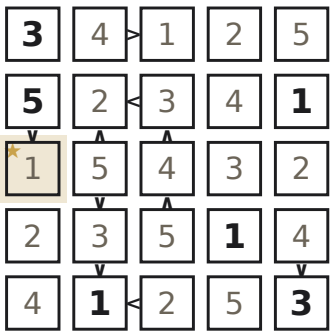
Day 21 · star = 1

	1	**2**		**3**	**4**	
	5	2	4	3	1	
	4	★1	5	2	**3**	
2	3	5	**2**	1	4	**2**
4	2	**3**	1	4	5	**1**
	1	4	**3**	**5**	2	**2**
		2				

Day 22 · star = 3

6	2	**4**	1	3	5
3	**1**	**5**	**2**	4	6
1	★3	2	6	**5**	4
5	4	6	**3**	2	1
4	6	3	5	1	**2**
2	5	**1**	**4**	6	**3**

Day 23 · star = 1



Day 24 · star = 5

		3				
	4	3	★5	**1**	2	**2**
	1	4	3	2	5	
1	5	1	2	4	3	
	2	5	4	3	1	**4**
2	3	2	**1**	**5**	**4**	**2**
					2	

Thank you for playing

Every puzzle in this calendar was generated and verified by computer to have exactly one solution, and the clue chain was checked so that a single suspect survives Day 24. If you enjoyed the case, a review on Etsy helps a small studio more than you know.

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